A Bijection and a Weighted Series

Sarbajit Manna

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Problem

Does there exist a bijective function

$$f : \mathbb{N} \to \mathbb{N}$$

such that

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^2} < +\infty?$$

Solution

Initial Approach

Since $f : \mathbb{N} \to \mathbb{N}$ is required to be bijective, a natural first choice is the identity map

$$f(n) = n.$$

This map is injective, since

$$f(n_1) = f(n_2) \implies n_1 = n_2,$$

and it is surjective because every $m \in \mathbb{N}$ is the image of m. Hence $f(n) = n$ is a bijection from $\mathbb{N}$ onto itself.

For this choice,

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^2} = \sum_{n=1}^{\infty} \frac{n}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n}.$$

The last series is the harmonic series, and therefore diverges.

My initial intuition was that the identity permutation should give the smallest possible value of the series among all bijections $f : \mathbb{N} \to \mathbb{N}$. Since even this choice gives the divergent harmonic series, I was led to the conclusion that no bijection can make the given series converge.

Gap in the Initial Argument

The preceding argument contains an unjustified step. Although $f(n) = n$ gives the harmonic series, this alone does not show that the identity permutation minimizes

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^2}$$

over all bijections $f : \mathbb{N} \to \mathbb{N}$.

In particular, bijectivity does not imply the pointwise inequality

$$f(n) \geq n.$$

Thus the intuition behind the initial approach requires a separate argument before the conclusion can be established rigorously.

Proof

Suppose that

$$f : \mathbb{N} \to \mathbb{N}$$

is a bijection. We show that

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^2}$$

must diverge.

For $N \in \mathbb{N}$, let

$$S_N = \sum_{n=1}^{N} \frac{f(n)}{n^2}$$

and denote the N-th harmonic number by

$$H_N = \sum_{n=1}^{N} \frac{1}{n}.$$

We write

$$\frac{1}{n} = \frac{\sqrt{f(n)}}{n} \frac{1}{\sqrt{f(n)}}.$$

Therefore, by the Cauchy–Schwarz inequality,

$$\left(\sum_{n=1}^{N} \frac{1}{n} \right)^2 \leq \left(\sum_{n=1}^{N} \frac{f(n)}{n^2} \right) \left(\sum_{n=1}^{N} \frac{1}{f(n)} \right).$$

Hence

$$H_N^2 \leq S_N \sum_{n=1}^{N} \frac{1}{f(n)}.$$

Since f is injective, the numbers

$$f(1), f(2), \ldots, f(N)$$

are N distinct positive integers. Among all sets of N distinct positive integers, the sum of the reciprocals is maximized by $\{1, 2, \ldots, N\}$. Consequently,

$$\sum_{n=1}^{N} \frac{1}{f(n)} \leq \sum_{n=1}^{N} \frac{1}{n} = H_N.$$

It follows that

$$H_N^2 \leq S_N H_N.$$

Since $H_N > 0$, division by H_N gives

$$S_N \geq H_N.$$

Thus, for every $N \in \mathbb{N}$,

$$\boxed{\sum_{n=1}^{N} \frac{f(n)}{n^2} \geq \sum_{n=1}^{N} \frac{1}{n}.}$$

But the harmonic series diverges, so

$$H_N \longrightarrow +\infty \qquad \text{as } N \to \infty.$$

Therefore

$$S_N \longrightarrow +\infty,$$

and hence

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^2} = +\infty.$$

Thus there does not exist a bijective function $f : \mathbb{N} \to \mathbb{N}$ for which

$$\sum_{n=1}^{\infty} \frac{f(n)}{n^2} < +\infty.$$

$\square$

Problem source. mmathemagics, Instagram, 27 July 2026.